

ECSE 4660/6660: Internetworking of Things

Lecture 6: PHY Layer Fundamentals – Part III

Ish Jain
ECSE, RPI

Spring 2026

Administrative

- Use Piazza for Lab questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
 - Asking Questions by email or Webex is not recommended
 - Piazza allows peer-discussion and faster response (can even post anonymously)
 - 5% Class Participation grades
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Project group formation **due 1/30!!** }
 - Add Project Group here [IoT_SP26_project.xlsx](#)
- Project Proposal submission **due 2/6**
 - Template Provided
 - Rubrics Provided
- HW1 on **LMS** **due 2/6** *fr*
 - *friday*
 - *today's lec*
- Lab is located in JEC 5312
 - Access is provided to all registered students
 - Collect Lab-kit (Rpi etc) during TA hours
 - You can visit lab anytime Mon-Fri 7AM-10PM
 - Labs are done **individually**-Plagiarism penalized heavily (See Syllabus pdf)
 - TA hours ~~4-5~~ PM daily in 5312 lab *3:30-4:30*
 - Lab 1 extended due **Mon, 2/2** *today*
 - Lab 2 due **Fri, 2/13** *2 weeks*

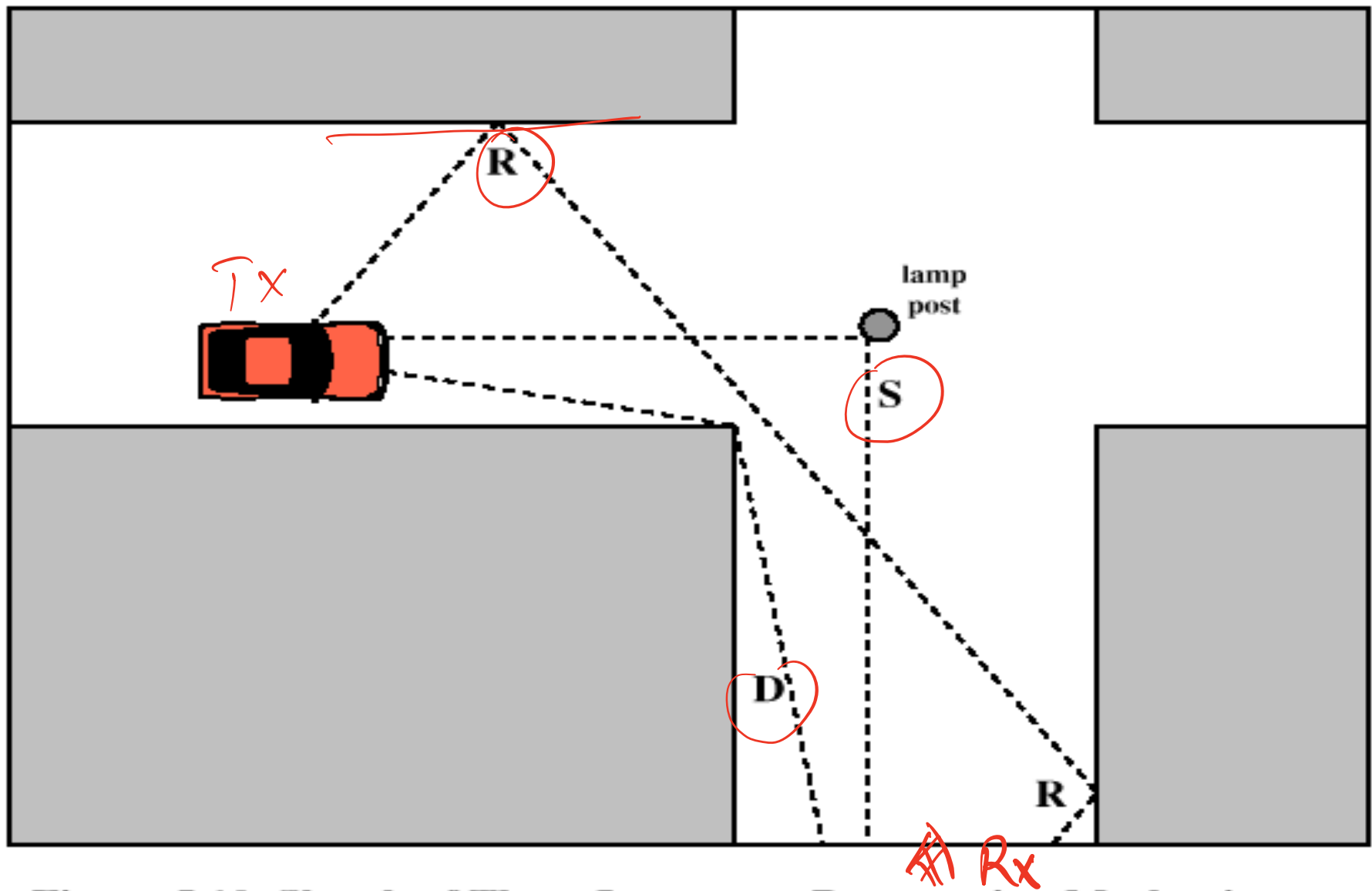
References

- Reference for Part III:
- B. Carlson, Communication Systems, McGraw Hill, Chapters 11-12, 3rd Ed.
- V. Garg, Wireless Communications & Networking, Morgan Kauffman, 2007. (available as e-book for RPI students)
 - Chapter 4: An Overview of Digital Communication Transmission
 - Chapter 9: Modulation Schemes

Wireless Transmission: Outline

1. Multi-path Fading
2. Modulation Schemes (Bits to Symbol)
3. Error Detection and Correction

Multipath Propagation



**Figure 5.10 Sketch of Three Important Propagation Mechanisms:
Reflection (R), Scattering (S), Diffraction (D) [ANDE95]**

The Effects of Multipath Propagation

■ Fading

(1)

- Multiple copies of a signal may arrive at different phases
- If phases add destructively, the signal level relative to noise declines, making detection more difficult

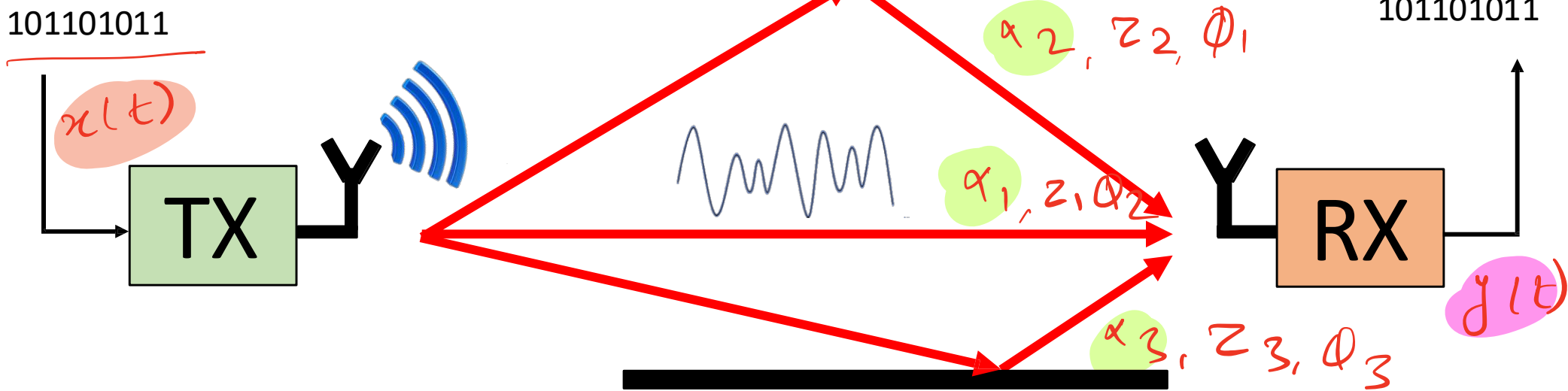
■ Inter-Symbol Interference (ISI)

(2)

- One or more delayed copies of a pulse may arrive at the same time as the primary pulse *for a subsequent bit*

Multipath Channel

metal - strong
wood - weak



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2) + \alpha_3 e^{j\phi_3} x(t - \tau_3) \dots$$

$$y(t) = \sum_k \alpha_k e^{j\phi_k} x(t - \tau_k) = \sum_k h_k(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

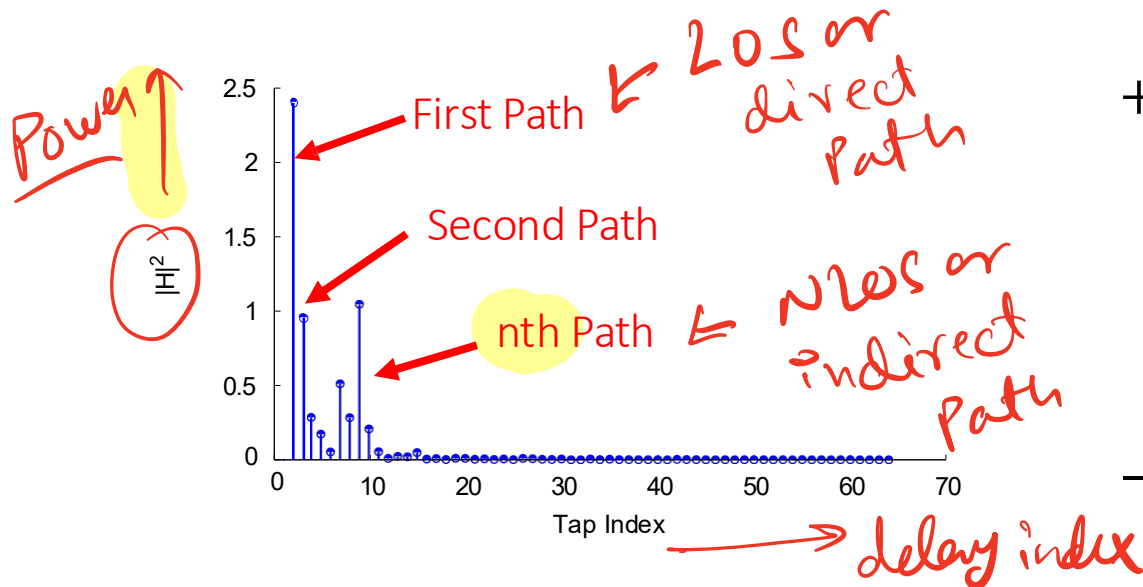
↑
convolution

$h(t)$ is channel impulse response.

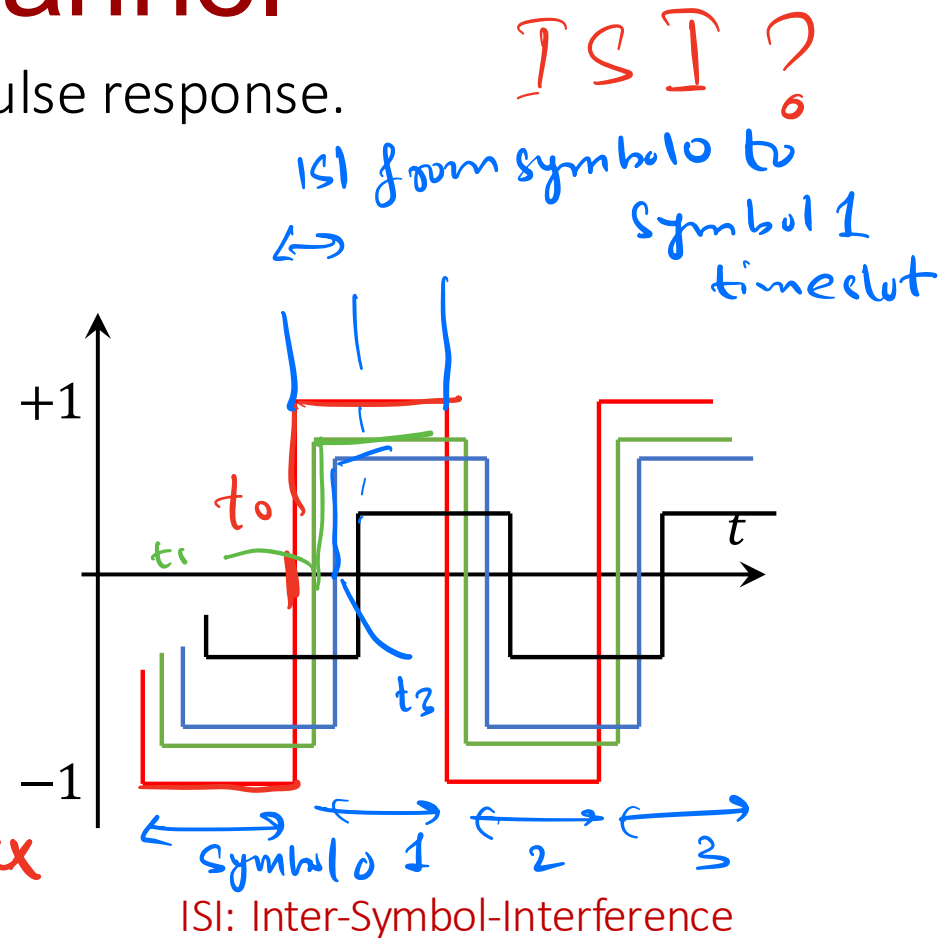
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = \boxed{h(t) * x(t)}$$



Multi-tap Channel



Symbols arriving along late paths interfere with following symbols.

How to mitigate ISI

$\Rightarrow$ increase symbol duration

But, it compromises bit rate

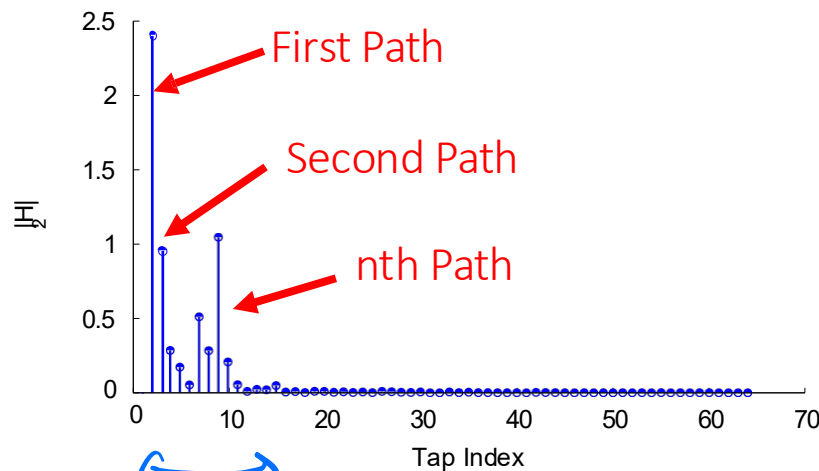
$\Rightarrow$ OFDM

$\Rightarrow$ MCS class

Multipath Channel

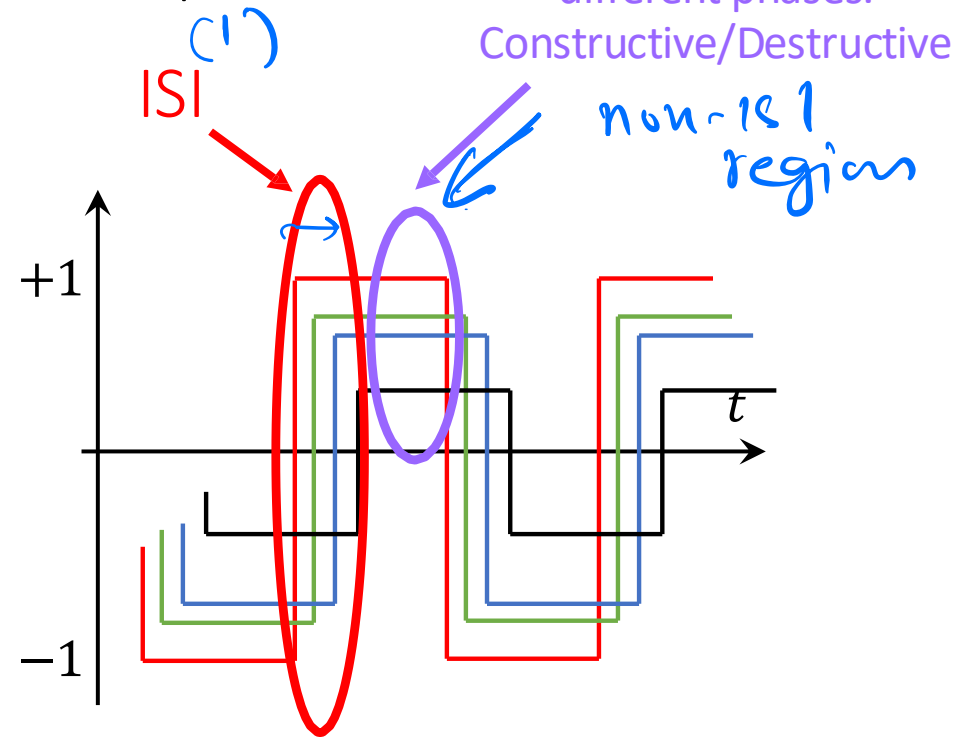
$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



ns.

Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Channel Fading

Symbols arriving along different paths sum up destructively

$\frac{10m}{3e8}$, $\frac{20m}{3e8}$
 $\Rightarrow 33ns$, $66ns$

Fading – constructive and destructive Interference

$$y(t) = \sum_{k=1}^L \alpha_k e^{j\phi_k} x(t - \tau_k)$$

↑ # of paths

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2)$$

$\downarrow$ FFT $\downarrow$ FFT $\downarrow$ FFT

$$Y(f) = \alpha_1 e^{j\phi_1} \underbrace{\text{FFT}(x(t - \tau_1))}_{x(f) e^{j2\pi f \tau_1}} + \alpha_2 e^{j\phi_2} \text{FFT}(x(t - \tau_2))$$

$$= \alpha_1 e^{j\phi_1} x(f) e^{j2\pi f \tau_1} + \alpha_2 e^{j\phi_2} x(f) e^{j2\pi f \tau_2}$$

$$Y(f) = X(f) \left[\begin{aligned} & x_1 e^{j\phi_1} e^{j2\pi f z_1} \\ & + x_2 e^{j\phi_2} e^{j2\pi f z_2} \end{aligned} \right]$$

$$Y(f) = X(f) \times H(f)$$

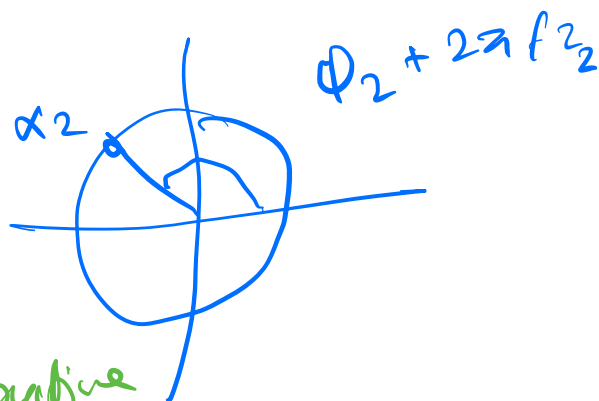
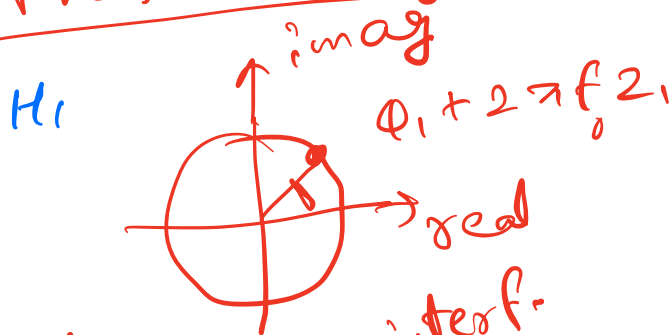
fix some freq to

two components

$$H_1 = x_1 e^{j\phi_1 + j2\pi f_0 z_1}$$

$$H_2 = x_2 e^{j\phi_2 + j2\pi f_0 z_2}$$

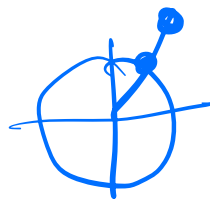
Phaser diagram



deep fade or destructive interf.

$$x_1 = x_2 \angle H(f)$$

constructive interference



OFDM

“Fading”

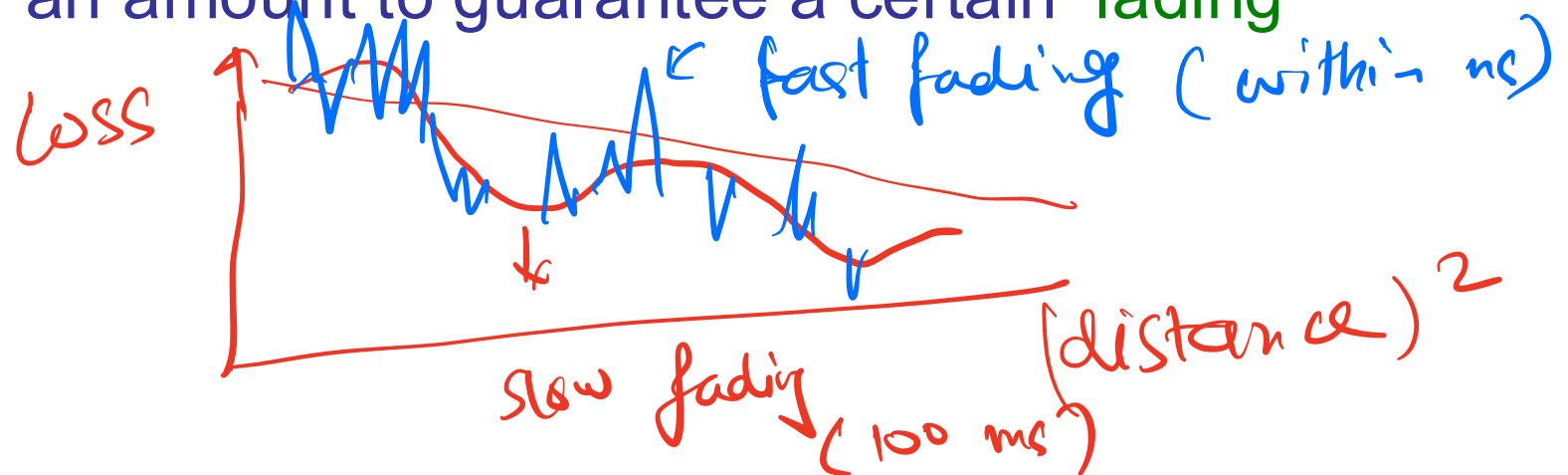
- Fading is the change in received power of a signal over time due to multi-path.
- Types of Fading
 - ✓ ■ Slow fading ← obstacles
 - ✓ ■ Fast fading ← ISI, multipath
 - Rayleigh fading ← lots of NLOS (weak LOS)
 - Rician fading ← strong LOS + weak NLOS
 - ✓ ■ Flat fading ← main LOS (non LOS)
 - ✓ ■ Frequency-Selective fading ← due to multipath

Slow vs Fast Fading

- In urban (or non free-space) environments, the two common kinds of 'fading' may take place
 - Slow fading: typically due to the movement of obstacles or transmitter/receiver, causing slow variation in the signal strength
 - Fast Fading: due to multipath propagation

Shadow (slow) fading

- The received signal strength from the transmitter will be different – i.e. it will vary around the mean value given by the path-loss equation.
- It is called 'slow' because the variations are typically much slower than 'fast fading'
- Several measurements and simulations indicate that this variation can be expressed as a log-normal random variable
- It is typically compensated for by increasing the transmit power by an amount to guarantee a certain 'fading margin'



Multi-path (fast) fading

- A signal may be received at the receiver through different 'paths' creating copies with different phase.
- Fast fading is the fluctuation of the signal amplitude because of the addition of signals arriving with different 'phases'
- The phase of the arriving paths changes rapidly, causing fast fluctuations of the signal amplitude ↑ no scale
- The distribution of the amplitude follows:
 - ■ Rayleigh distribution: if all signals suffer nearly the same attenuation, but arrive with different phases, and no distinct dominant path (outdoor)
 - ■ Rician distribution: when there is a strong LOS signal component (indoor)
- Mitigation:
 - Increasing signal power is not very helpful
 - ✓ ■ Error control coding with interleaving
 - ✓ ■ Diversity schemes ← time, freq. } antennas } MIMO
 - Directional Antennas (reducing the number of multiple paths) ↓ MCS

Multipath Delay Spread (ISI)

- If the multiple paths also introduce **delays** that are comparable to (or larger than) the symbol duration, the received waveform **spreads** into the neighboring symbols and produces ISI (irreducible error rates).
- Several models exist for multipath delay spread
- Mitigation: (we will study these later)
 - Equalization ← *estimate those delays / phases in channel & compensate for them*
 - 36 → ■ Direct Sequence Spread Spectrum
 - ■ Directional Antennas (reducing the multipath components)

$$\begin{aligned} y(t) &= x(t) * h(t) \\ Y(f) &= X(f) \times H(f) \end{aligned} \quad \left. \vphantom{\begin{aligned} y(t) &= x(t) * h(t) \\ Y(f) &= X(f) \times H(f) \end{aligned}} \right\} \frac{Y(f)}{\hat{H}(f)} = \hat{X}(f)$$

$\hat{H}(f) \leftarrow$ est. channel

Wireless Transmission: Outline



1. Multi-path Fading
2. Modulation Schemes (Bits to Symbol)
3. Error Detection and Correction

00110

Now how do we send information?

- A single-frequency signal that is always on at the same frequency and amplitude conveys no information
- To transmit info, **some aspect of the signal has to be changed** i.e. Signal has to be *modulated*.

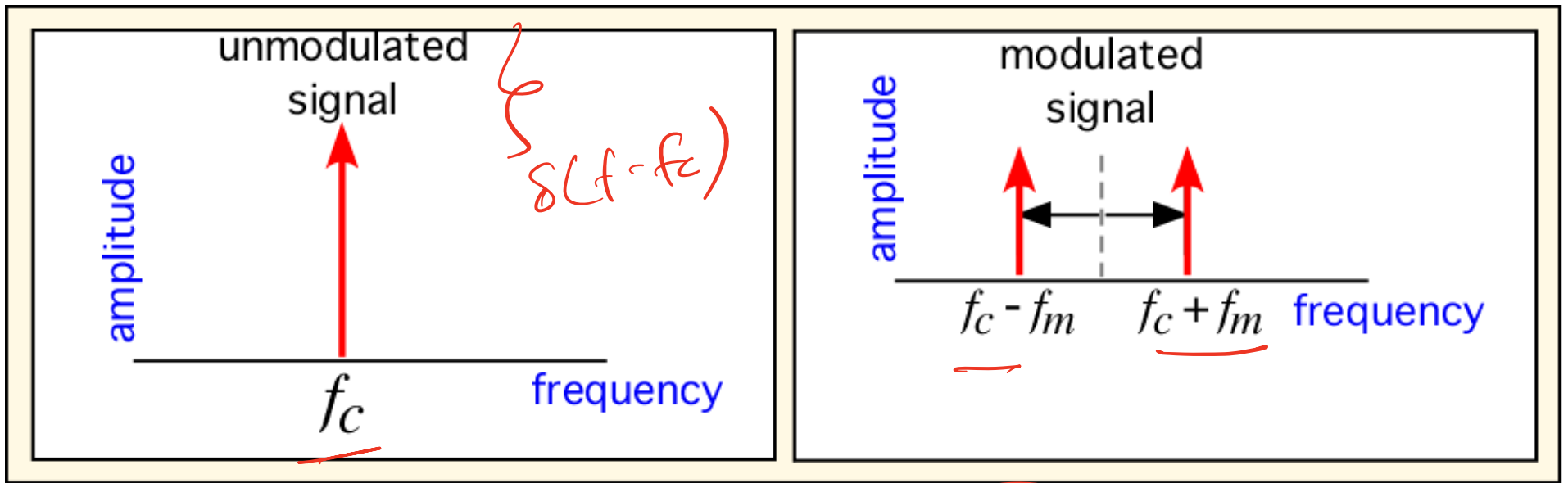
■ E.g.:

$$f(t) = m(t) \cos(\omega_c t)$$

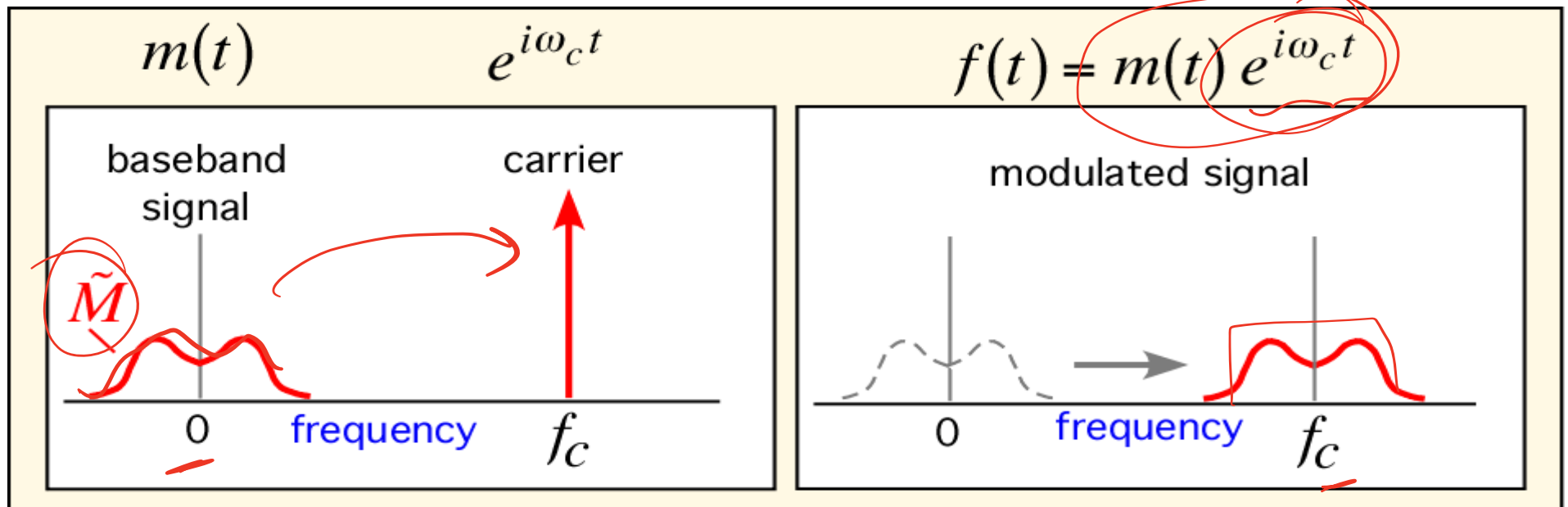
Modulated signal Modulating function Carrier frequency

$2\pi f_c$

$f_c = \text{carrier freq}$



$\cos(\omega_c t)$ modulated by $\cos \omega_m t$ $m(t) = \cos 2\pi f_m t$



$\cos(\omega_c t)$ modulated by a general $m(t)$ with Fourier transform $M(\omega)$

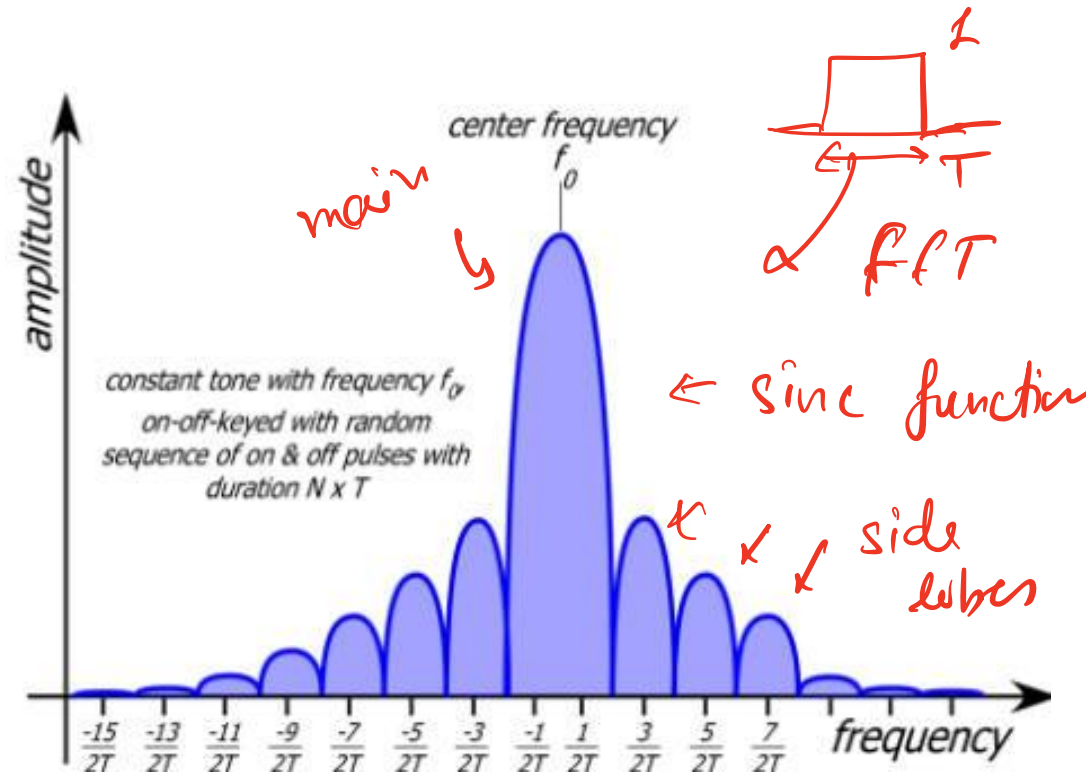
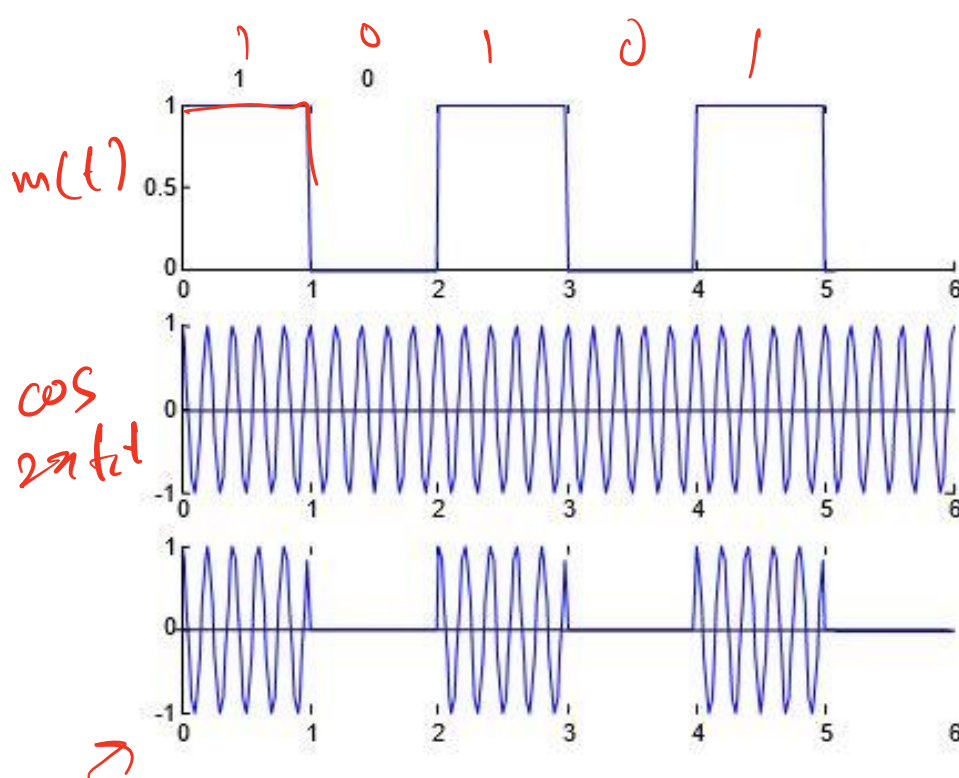
Digital Modulation

← good for IoT

($T = \text{duration of symbol}$)

- Map Bits to Signal Values
- On-Off Keying (ASK):

ASK = Amplitude shift key



$m(t) \cos 2\pi ft$

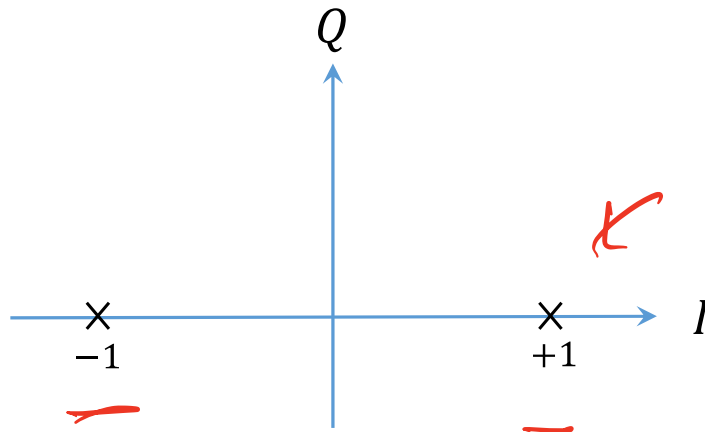
$\leftrightarrow \frac{1}{T} = \text{bandwidth}$

Digital Modulation

implemented in 4G+
or wifi
ad-hoc
systems

- BPSK : Binary Phase Shift Keying

PSK



$$\cos 2\pi f_c t$$

$$\cos(2\pi f_c t + \pi) = -\cos(2\pi f_c t)$$

Requires phase modulation (not amplitude modulation)

power P

0 $\rightarrow$ +1 $\rightarrow$ +P
1 $\rightarrow$ -1 $\rightarrow$ -P

Digital Modulation

- PAM : Pulse Amplitude Modulation
 - Scale to maintain total average power

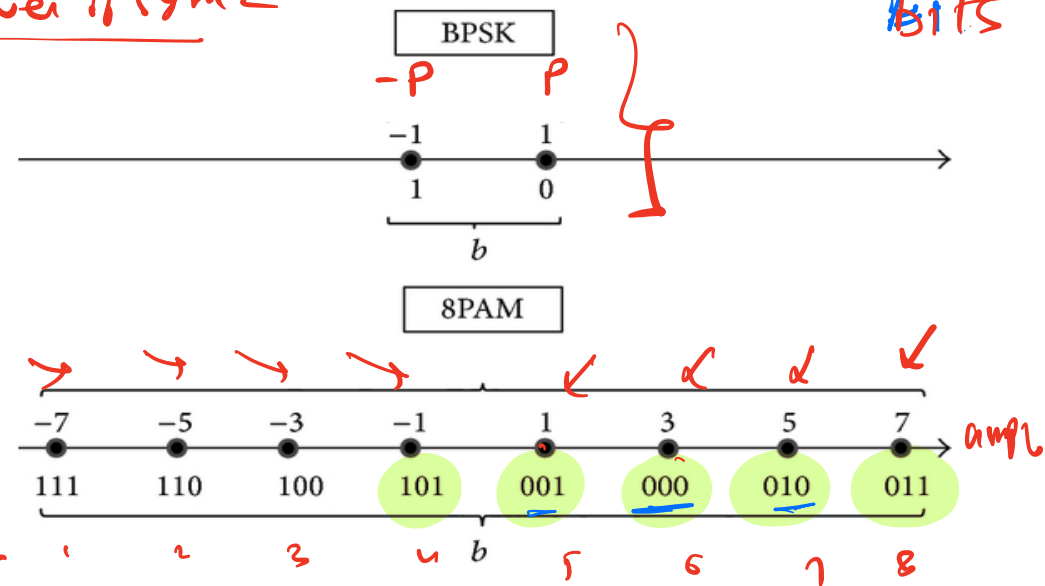
$$\begin{aligned} \text{Av. power} &= \frac{\text{Power of sym 1} + \text{Power of sym 2}}{2} \\ &= \frac{P + P}{2} \\ &= P \end{aligned}$$

$$\text{Av. Power} = \frac{\sum \text{Power of sym } i}{\text{H of symbols}}$$

$$\frac{2 \left[(1x)^2 + (3x)^2 + (5x)^2 + (7x)^2 \right]}{8} = P \quad \left(\begin{array}{l} -7x, -5x, -3x, -1x, 1x, 3x, 5x, 7x \\ \leftarrow \text{maintain the same total power} \end{array} \right)$$

$T_x \Rightarrow$ bits to symbol mapping

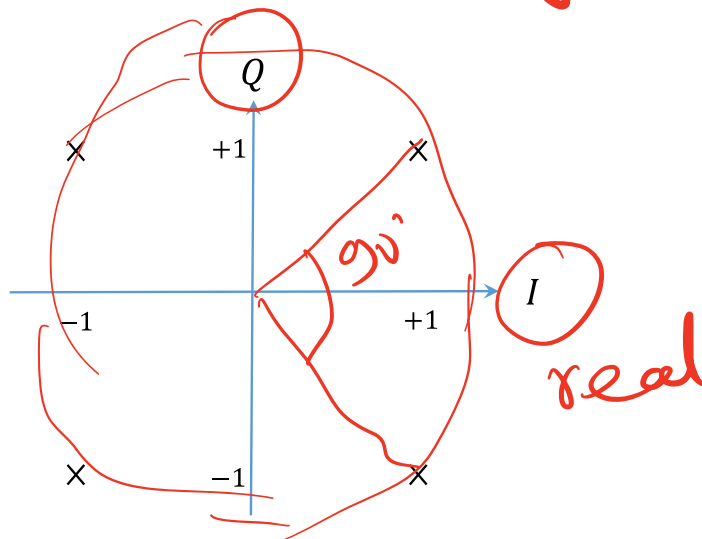
$R_x \Rightarrow$ symbol decoded $\Rightarrow$ map bits



Digital Modulation

- PSK : Phase Shift Keying

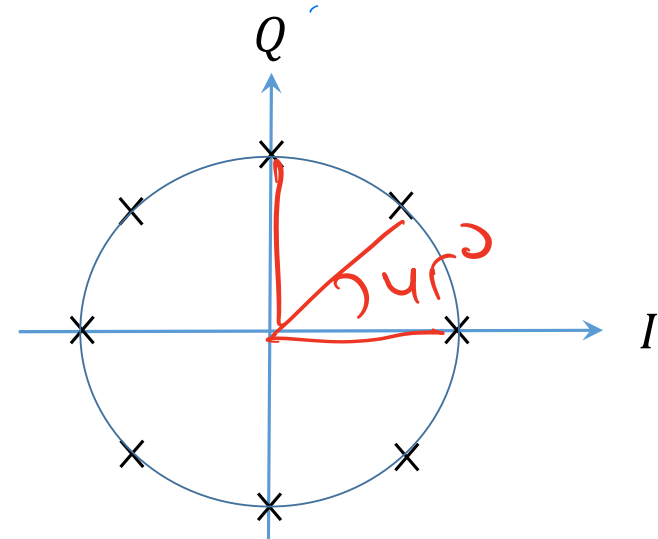
for imag.



4-PSK

QPSK : Quadrature Phase Shift Keying

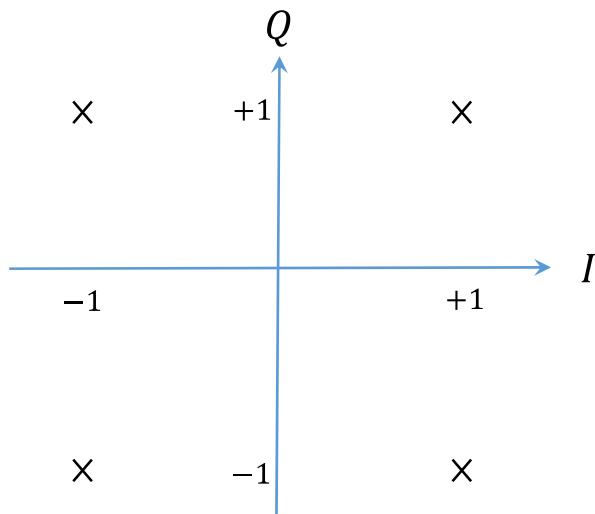
→ complex symbol



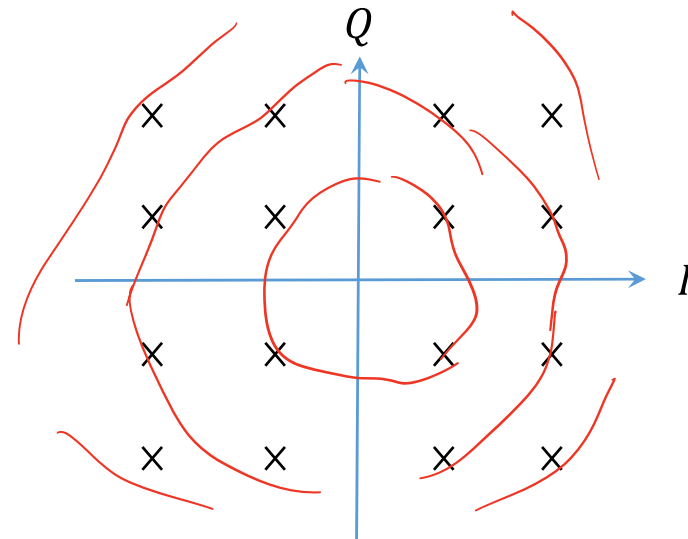
8-PSK

Digital Modulation

- QAM : Quadrature Amplitude Modulation



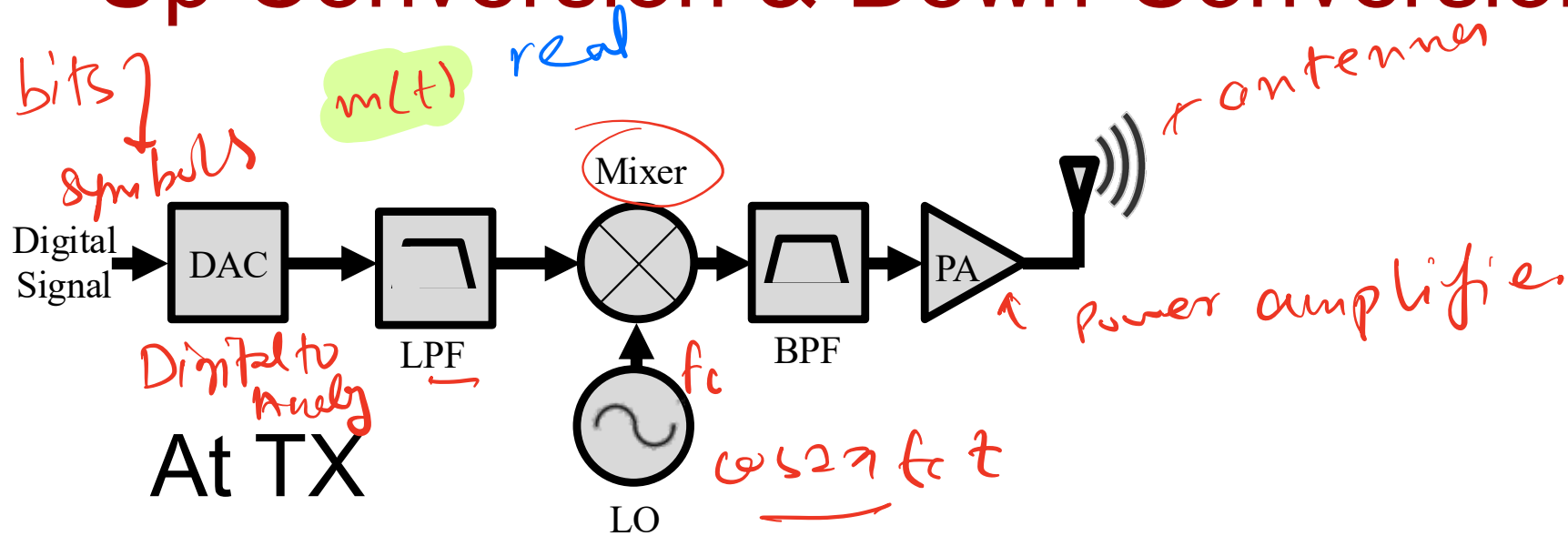
4 QAM $\Rightarrow$ QPSK



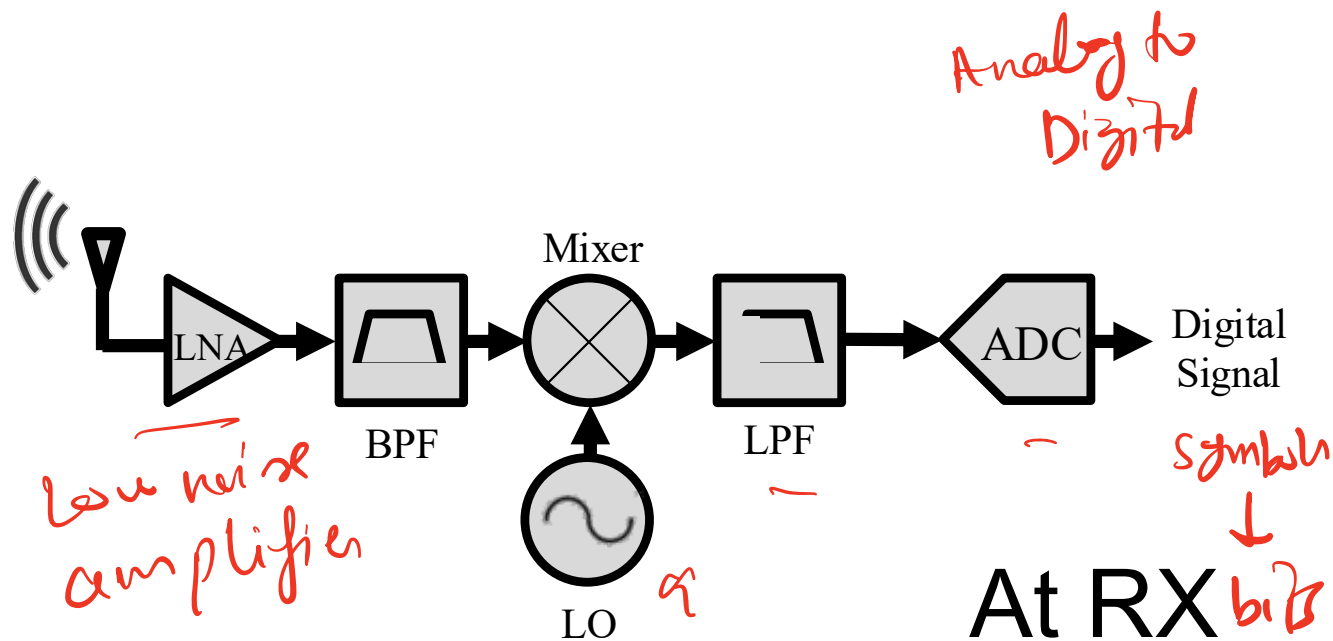
16 QAM

3 mag.
levels -

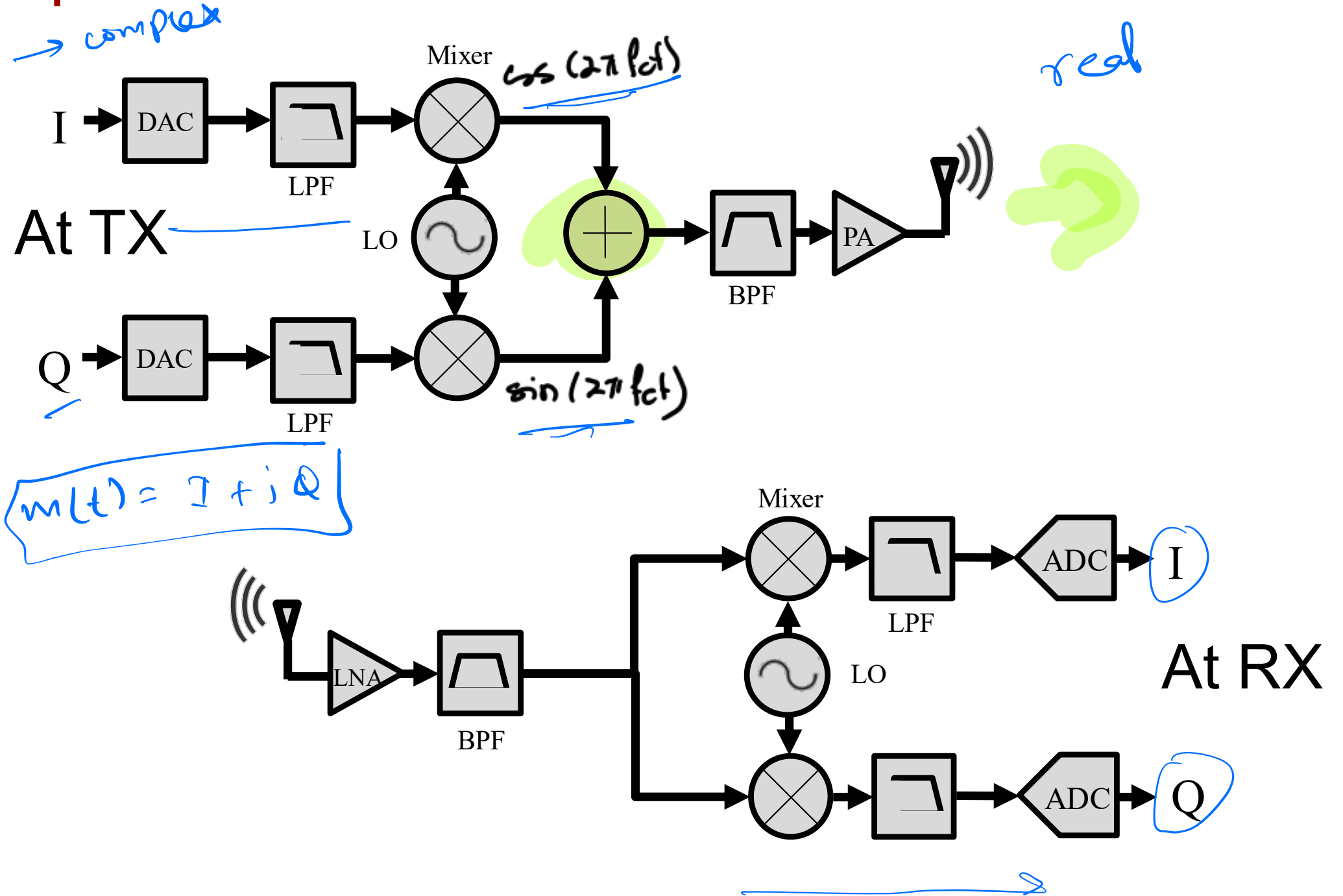
Up Conversion & Down-Conversion



real signal



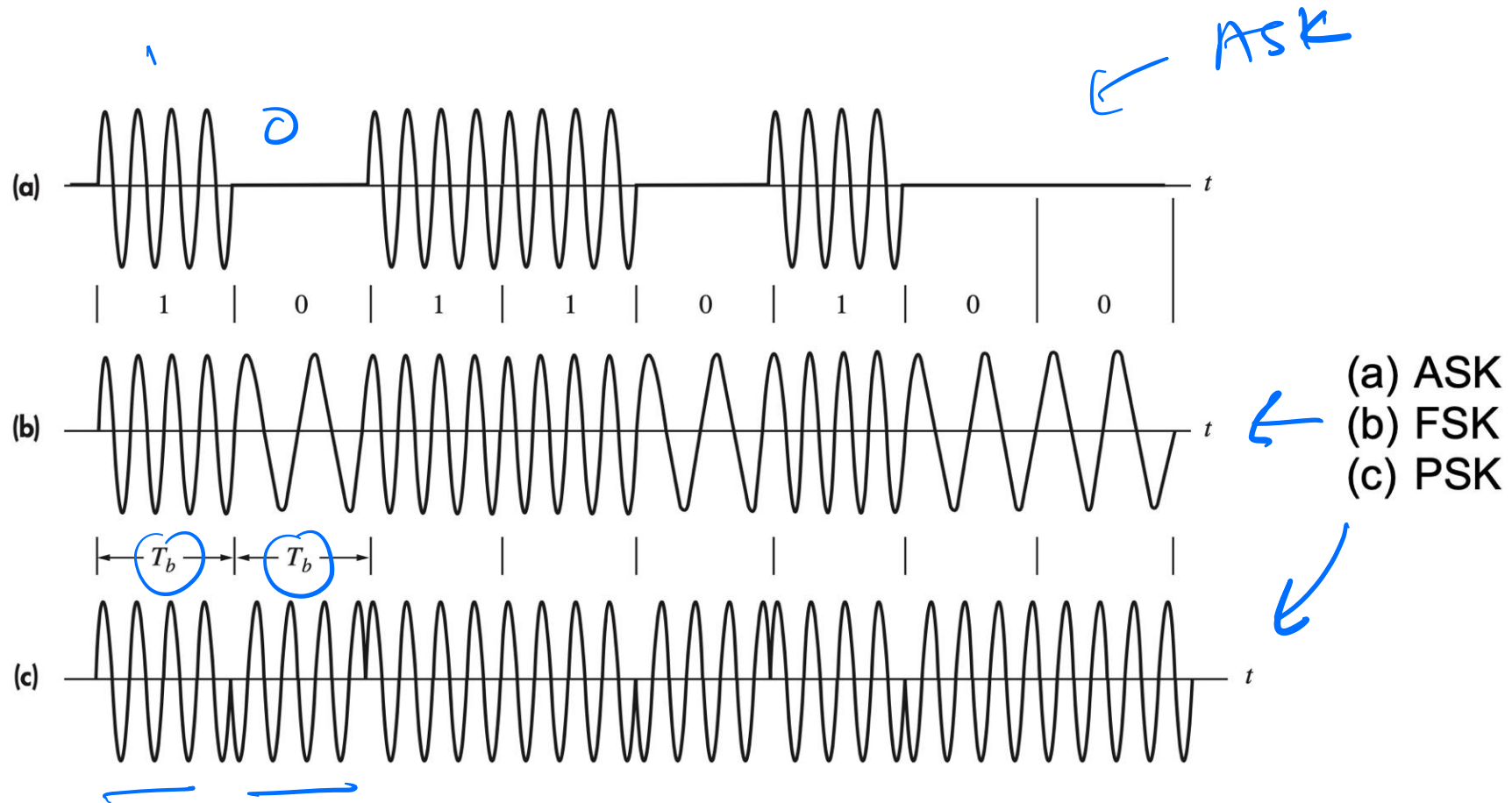
Up Conversion & Down-Conversion



To transmit a digital bit stream

- Convert the bit stream into the analog signal $a(t) \cos(\omega t + \theta)$. $m(t)$
- ASK: change $a(t)$ only, fix ω and θ
- FSK: change ω only, fix $a(t)$ and θ
- PSK: change θ only, fix $a(t)$ and ω
- QAM: change $a(t)$ and θ , fix ω

Example: ASK, FSK, PSK

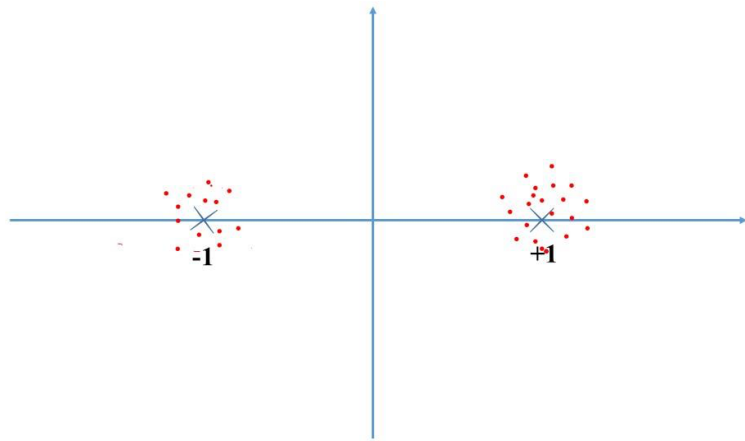


Wireless Transmission: Outline

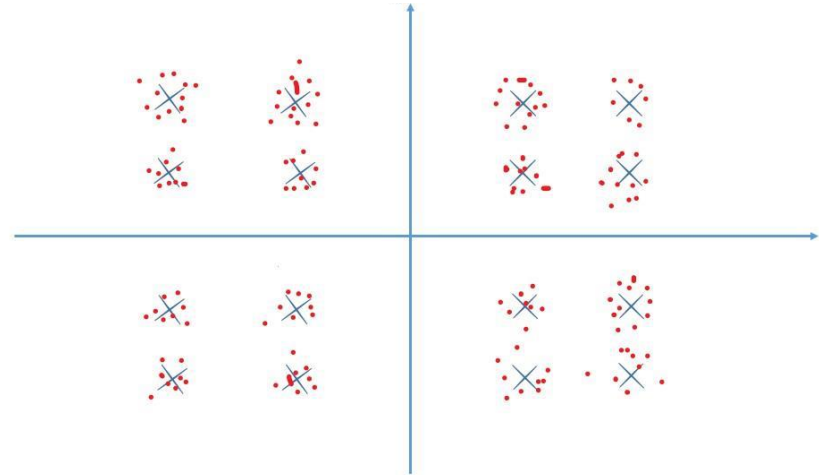
1. Multi-path Fading
2. Modulation Schemes (Bits to Symbol)
- 3. Error Detection and Correction**

Additive White Gaussian Noise

BPSK: 1 bit per symbol



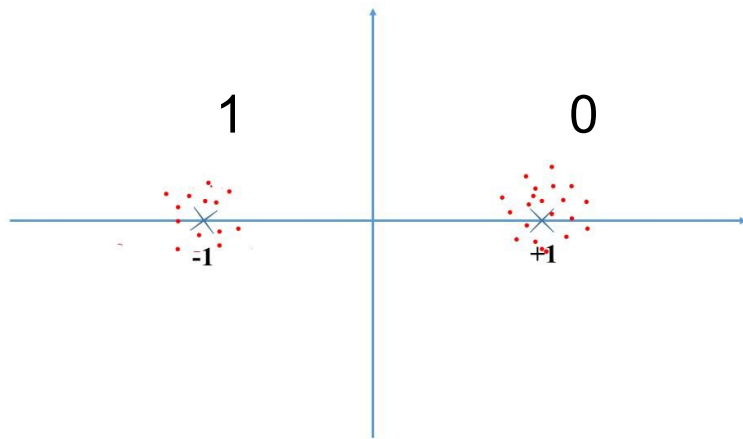
16 QAM: 4 bits per symbol



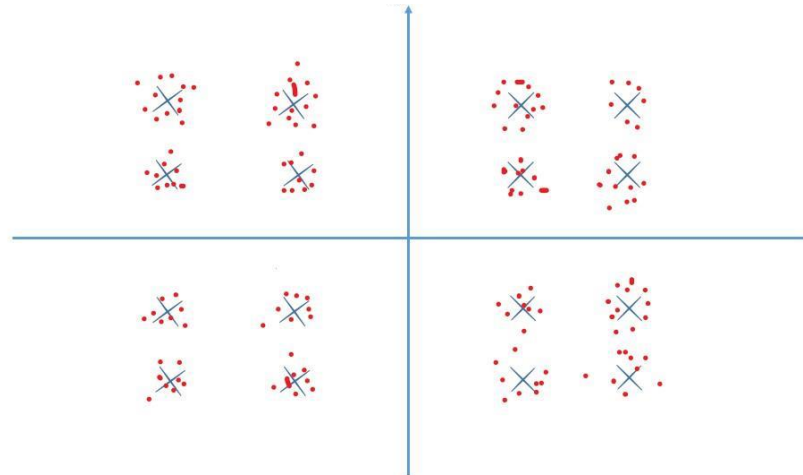
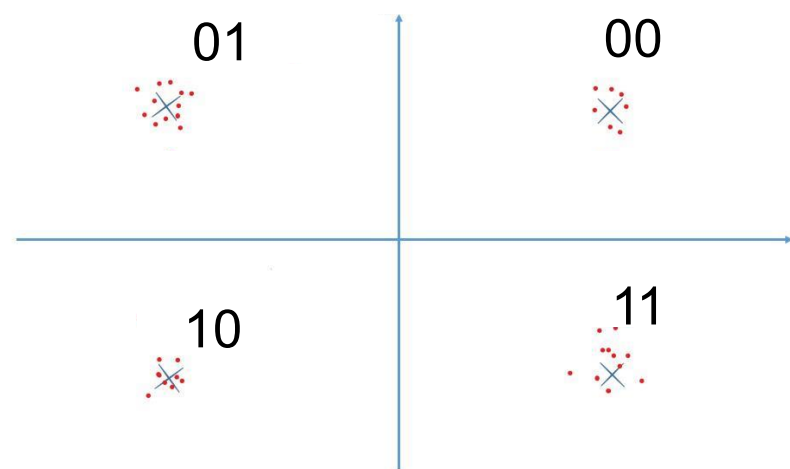
Maximum Likelihood Detector → Decision Boundary

How to map the bits to constellation points?

BPSK: 1 bit per symbol

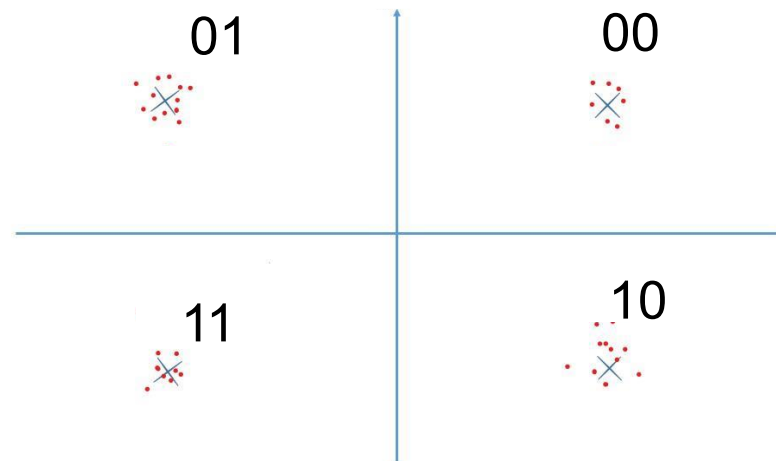
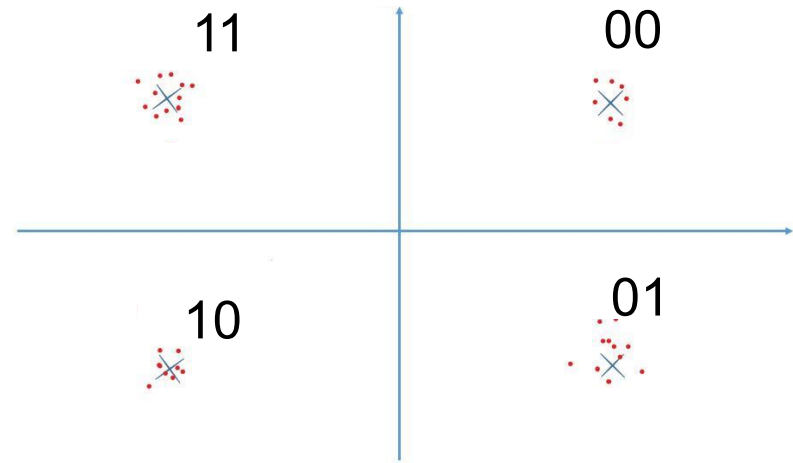
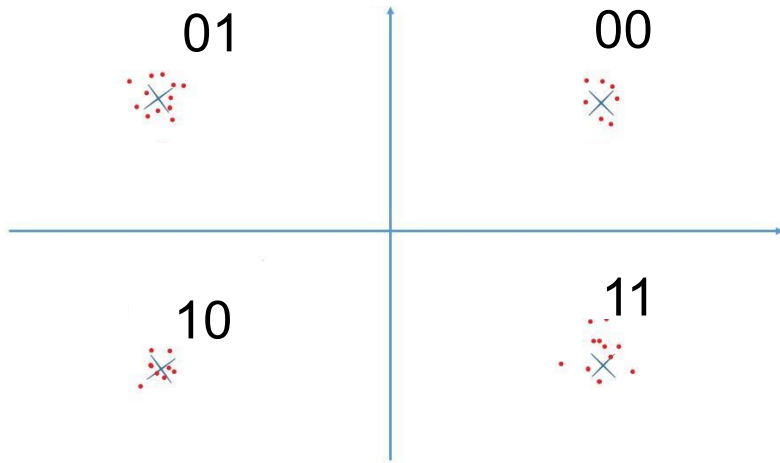


QPSK/4 QAM: 2 bits per symbol



How to map the bits to constellation points?

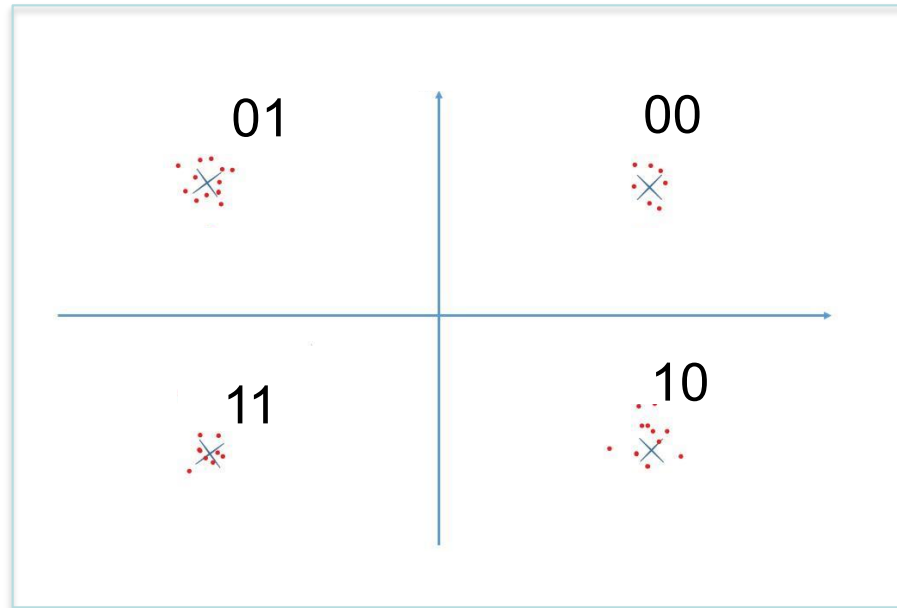
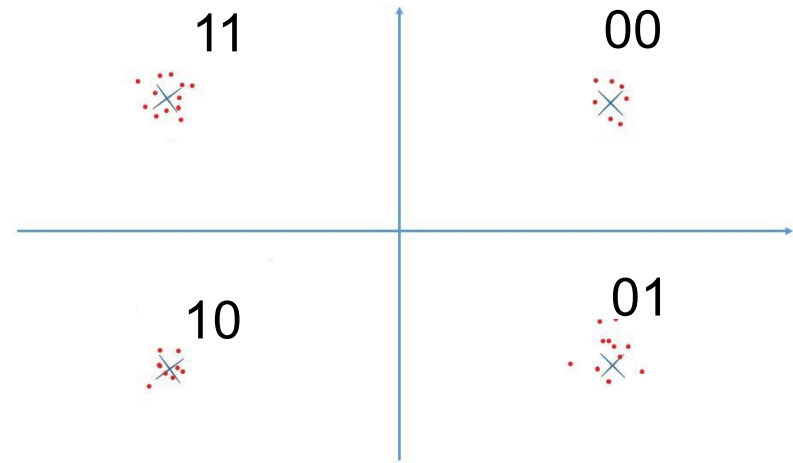
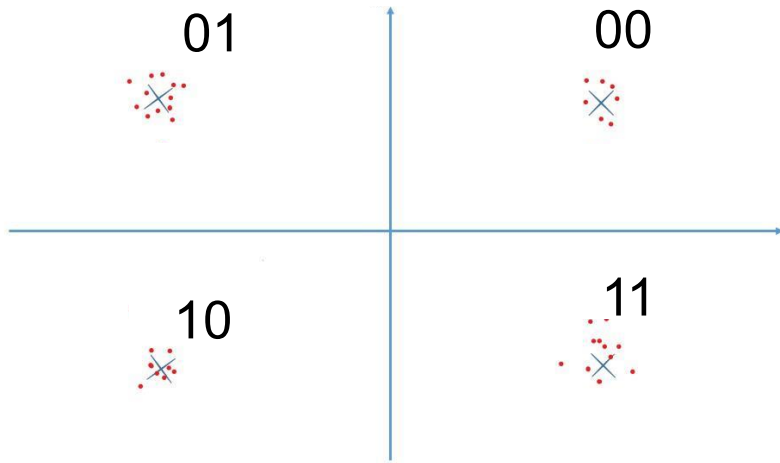
QPSK: 2 bits per symbol



Q: Which of the three mapping is better?

How to map the bits to constellation points?

QPSK: 2 bits per symbol



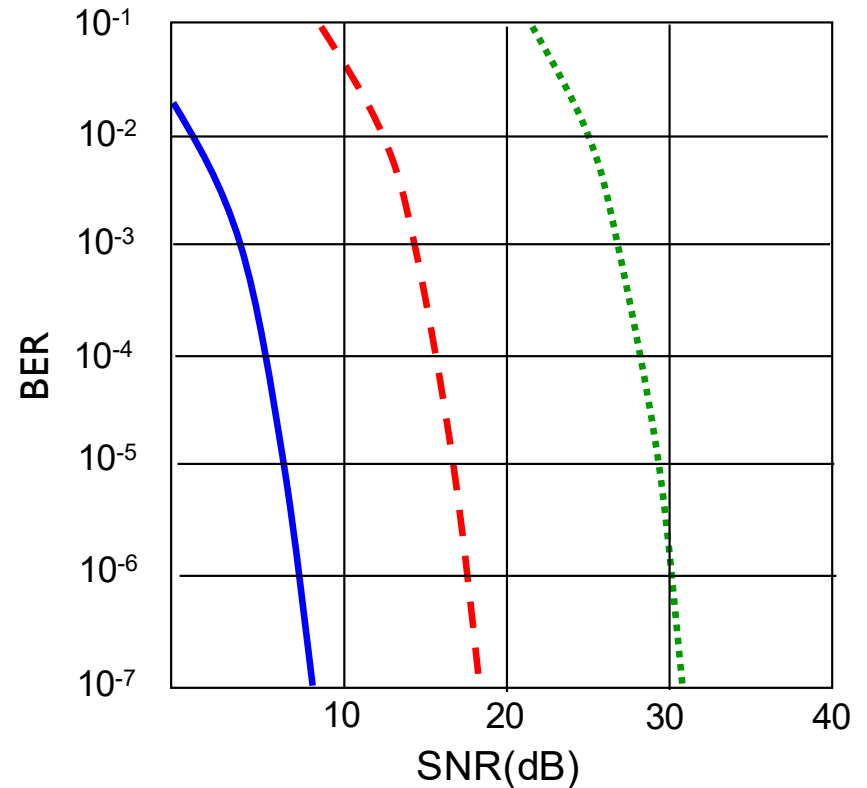
Gray Codes:

Minimize bit error

Adjacent symbols
only differ by 1 bit

Bit-Error-Rate

- *SNR versus BER tradeoffs*
 - *given physical layer modulation:*
Higher SNR $\rightarrow$ Low BER
 - *given SNR:* choose physical layer that meets BER requirement, giving highest throughput



..... QAM256 (8 Mbps)

- - - QAM16 (4 Mbps)

— BPSK (1 Mbps)

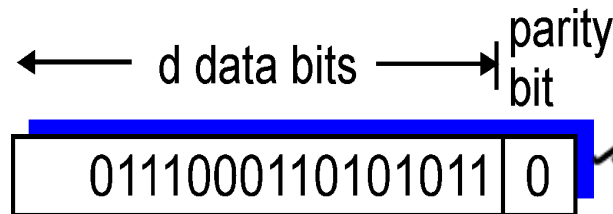
Higher Bit Rates

- From the previous example, higher bit rates could be achieved by:
 - Increasing the symbol rate (i.e. decrease T)
 - Nyquist criterion (to avoid aliasing): pulse rate must be less than $2W$.
 - Increase the number of possible symbols from 2 to $M > 2$ i.e. bit rate is $(\log M)/T$
 - How much can we “pack” while maintaining the same error probability per bit?
 - Need to increase the power levels (so that there is same degree of separation between the different levels)

Error Detection and Correction

- Add Redundant bit to
- Checksums → Detect Errors

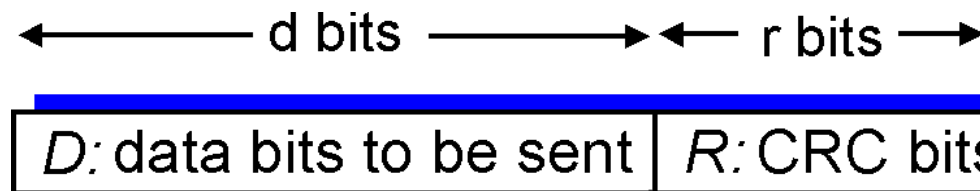
- Parity Check



even # of 1 $\Rightarrow 0$
odd # of 1 $\Rightarrow 1$

} 1 bit error
odd number
of bit flips

- CRC: Cyclic Redundancy Check



bit
pattern

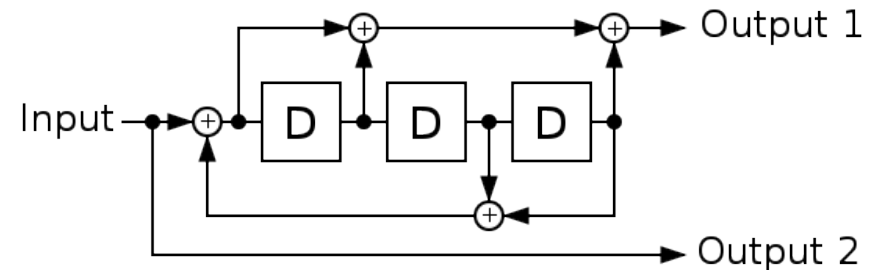
Larger r
 $\Rightarrow$ detect
more errors

Error Detection and Correction

■ FEC: Forward Error Correction

- Repetition Code
- Convolutional codes
- Reed Solomon codes

- Turbo codes
- LDPC codes



- Coding Rate = $\frac{2}{3}$ $\rightarrow$ every two bits add 1 more bit

$\uparrow$ coding rate $\Rightarrow$ less correct

$\downarrow$ coding rate $\Rightarrow$ redundancy $\Rightarrow$ $\uparrow$ correct bits

Lab 2

Open

File Edit View Run Tools Help

untitled - BPSK_SER_simulation

Run the flowgraph

Options
ID: BPSK_SER_simulation
Generate Options: QT GUI

QT GUI Range
ID: SNR
Label: SNR (dB)
Default Value: 10
Start: -15
Stop: 20
Step: 1

Variable
ID: modulation
Value: <gnuradio...a99cc4e0>

Variable
ID: BPSK_MAP
Value: -1, 1

Import
Import: math

Variable
ID: samp_rate
Value: 320k

Virtual Sink
Stream ID: GT

Random Source
Minimum: 0
Maximum: 2
Num Samples: 1k
Repeat: Yes

Chunks to Symbols
Symbol Table: -1, 1
Dimension: 1

Throttle
Sample Rate: 320k

Virtual Sink
Stream ID: Transmitter

Virtual Source
Stream ID: Transmitter

Add

Noise Source
Noise Type: Gaussian
Amplitude: 100m
Seed: 42

QT GUI Constellation Sink
Number of Points: 1.024k
Autoscale: No

Virtual Sink
Stream ID: Receiver

BPSK Decoder

Constellation Decoder
Constellation Object: ...0>

Virtual Source
Stream ID: Receiver

Virtual Source
Stream ID: GT

BER
Window Size: 1M

QT GUI Number Sink
Name: Bit Error Rate
Autoscale: No
Average: 0
Graph Type: Horizontal

Virtual Sinks and Virtual Sources with the same Stream ID are logically connected

Two blocks you need to write codes for

Core

- Audio
- Boolean Operators
- Byte Operators
- Channelizers
- Channel Models
- Coding
- Control Port
- Debug Tools
- Deprecated
- Digital Television
- Equalizers
- Error Coding
- FCD
- File Operators
- Filters
- Fourier Analysis
- GUI Widgets
- Impairment Models
- Instrumentation
- Level Controllers
- Math Operators
- Measurement Tools
- Message Tools
- Misc
- Modulators
- Networking Tools
- NOAA
- OFDM
- Packet Operators
- Pager
- Peak Detectors
- Resamplers
- Stream Operators
- Stream Tag Tools
- Symbol Coding
- Synchronizers
- Trellis Coding
- Type Converters
- UHD
- Variables
- Video
- Waveform Generators

<<< Welcome to GNU Radio Companion 3.7.13.4 >>>

Block paths:
/usr/local/share/gnuradio/grc/blocks
/usr/local/share/gnuradio/grc/blocks

Loading: "/home/uone/Gnuradio/gnuradioworkspace/RPI-IoT/BPSK_QPSK/Problems/BPSK_SER_simulation.grc"
>>> Done

running messages, including python code printout

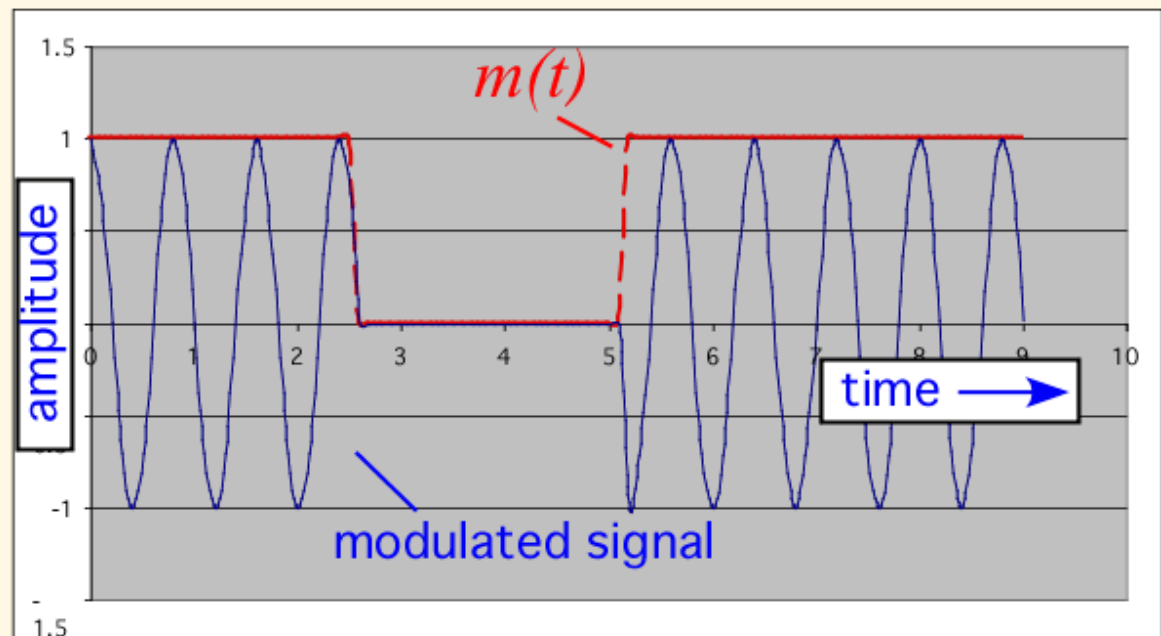
Id	Value
Imports	
import_0	import math
Variables	
BPSK_MAP	-1, 1
modulatic digital.constellation_bpsk()	base()
samp_rate	32e4
SNR	<Open Properties>

On Off Keying ($m(t)=0$ or 1)

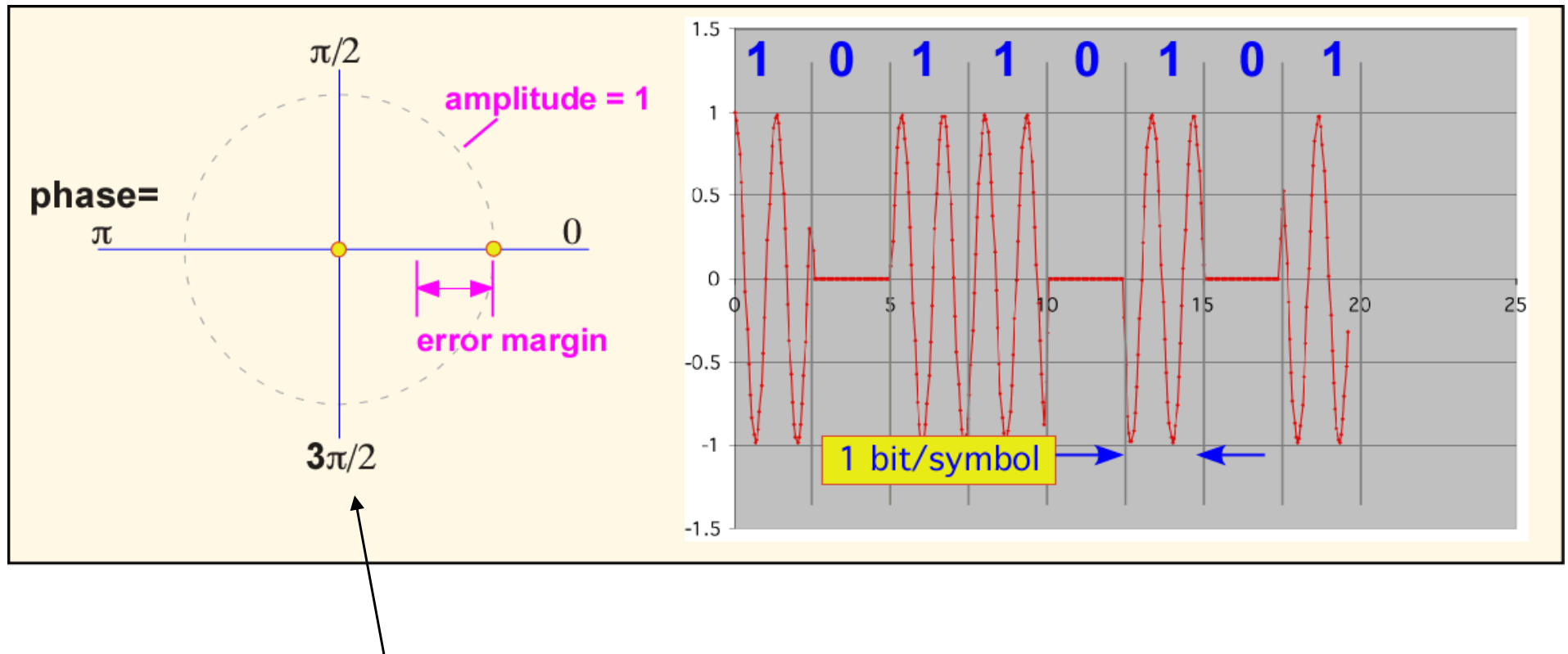
On-Off Keying

$$f(t) = m(t) \cos(\omega_c t)$$

$$m(t) = \begin{cases} +1 \\ 0 \end{cases}$$

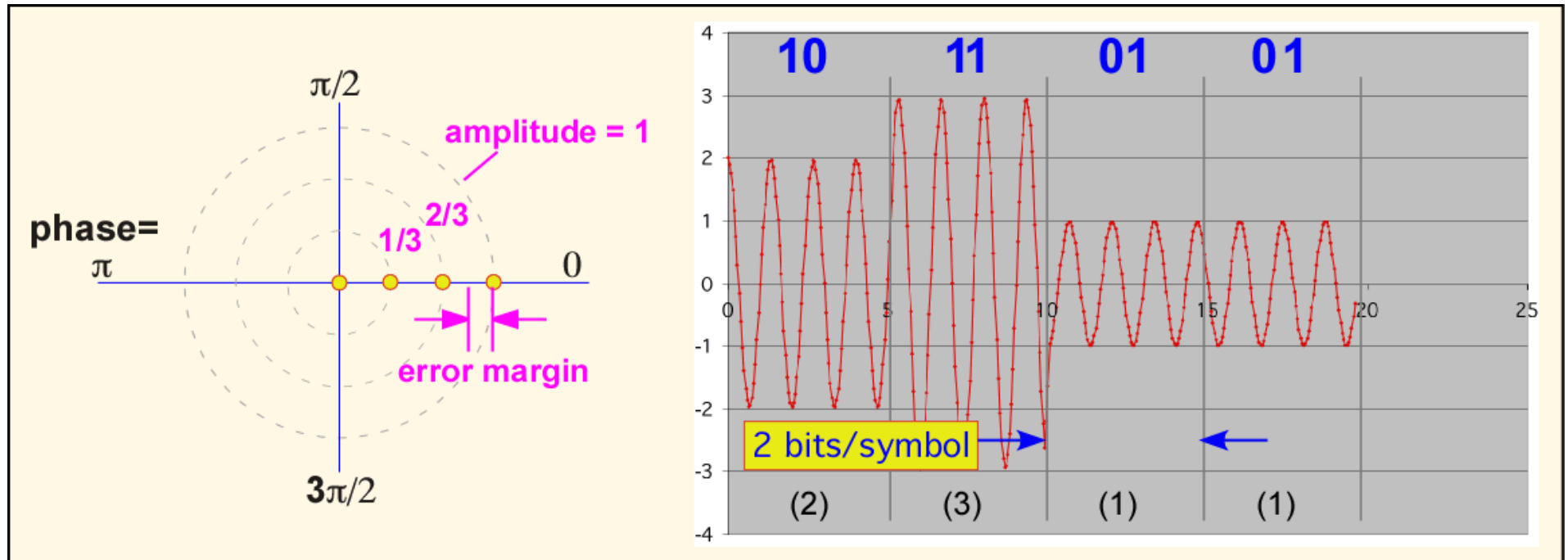


On Off Keying ($m(t)=0$ or 1)



Signal Constellation diagram to compute the probability of incorrect detection
(also called **Signal-Space** representation)

4-Amplitude Shift Keying

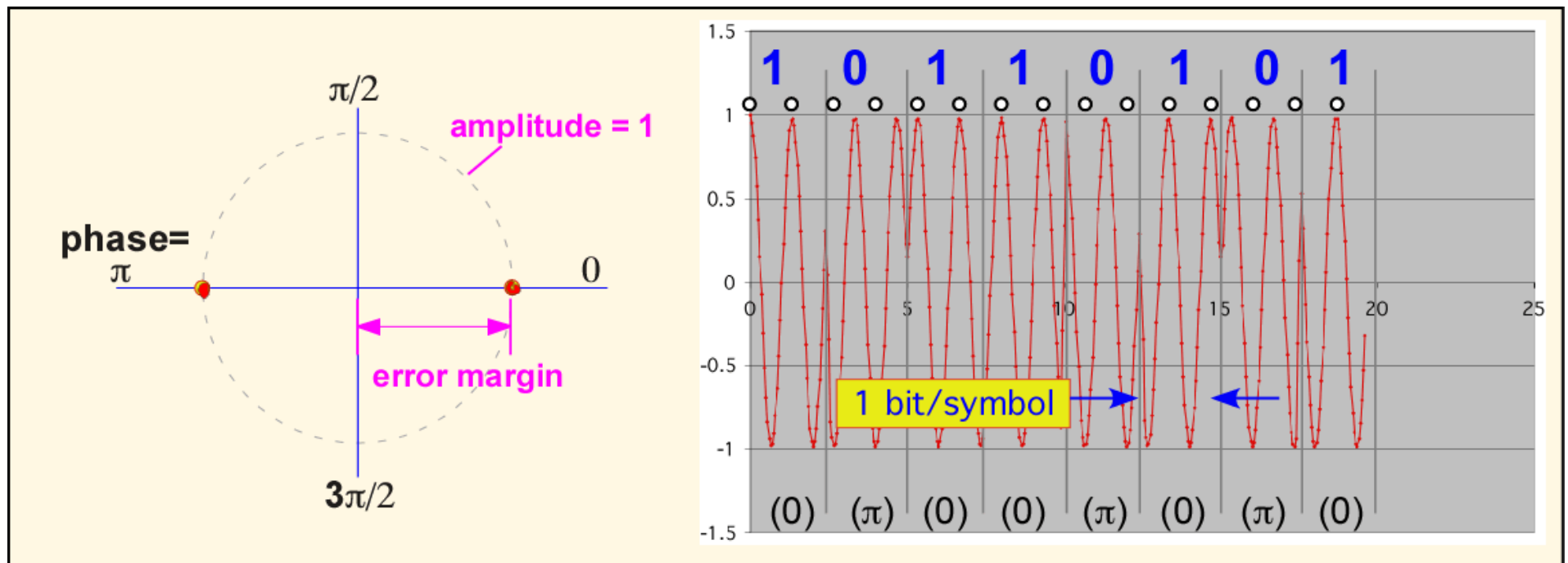


-4ASK transmits 2 bits per symbol

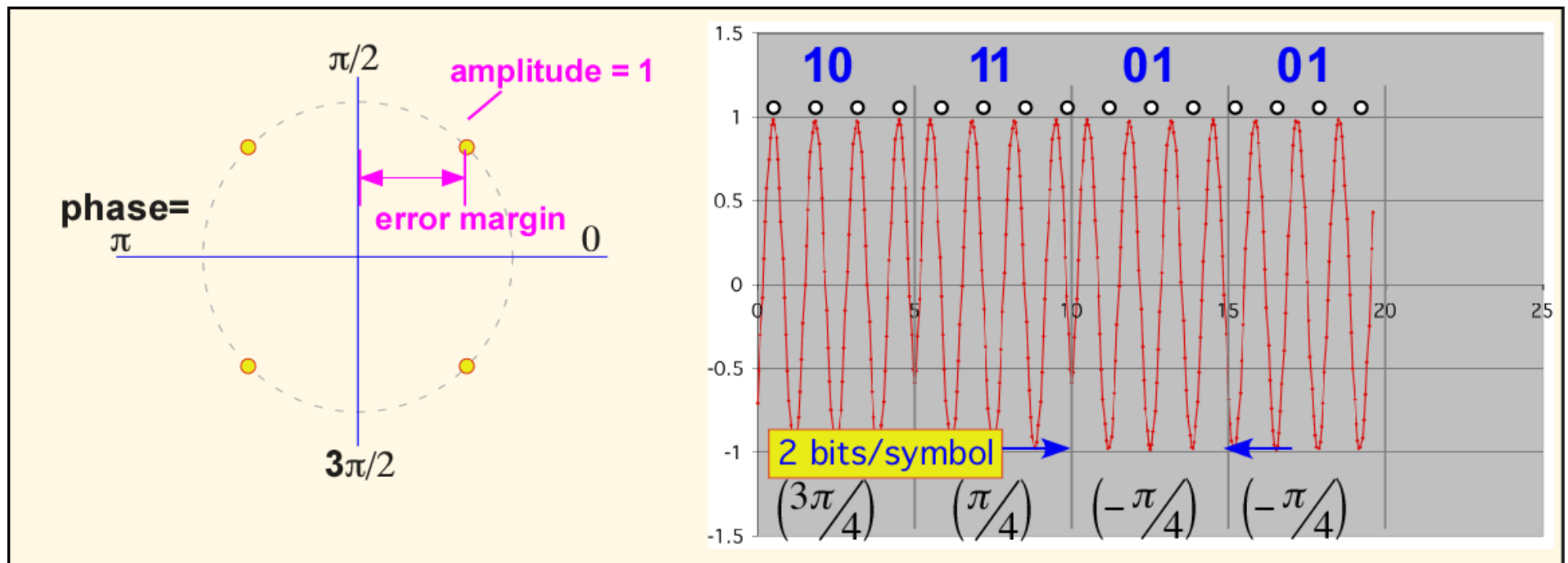
-Provides same data rate with half the bandwidth -(or, provides twice the data rate of OOK with same bandwidth)

-But, error margin is much smaller than OOK

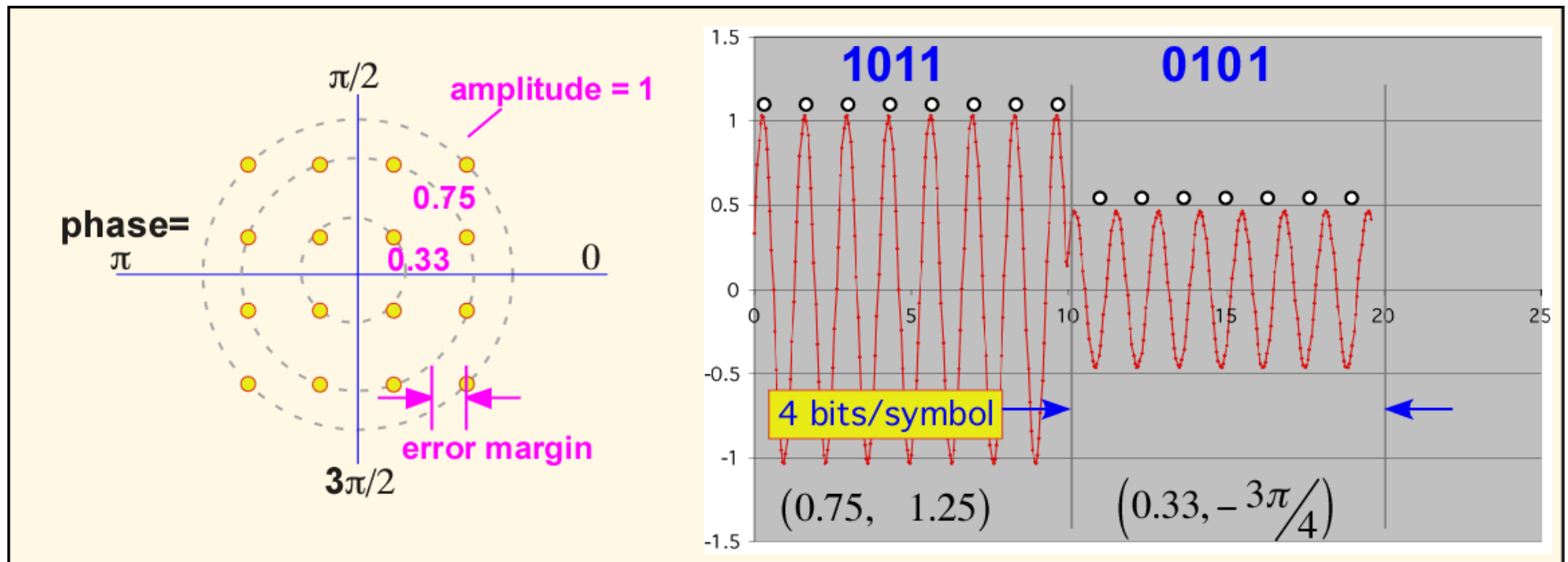
Binary Phase-Shift Keying (BPSK)



Quaternary Phase Shift Keying (QPSK)



16-Quadrature Phase Shift Keying (16QAM)



-A combination of amplitude and phase modulation, trying to keep the points in the constellation equally spaced from each other.